Secure File Sharing Using Blockchain

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Abstract— The process of safely sharing files has gained greater importance with the increased reliance of data sharing on digital systems, since the traditional models of systems are less secure, including unregulated systems, data breaches, and single point system failures. In this research, the methodology of designing distributed files sharing systems for the first time that use Inter-Planetary File System (IPFS) and blockchain systems are of primary concern . In the proposed system, blockchain is used to store the cryptographic hashes and IPFS to keep the files in a distributed manner [1],[5], allowing the system to be data immutable and to preserve its integrity[3], [4]. To help use the proposed model, Remix is used to deploy the smart contracts and Pinata is used for IPFS services. An improved model has better security, integrity, and transparency, which makes it a better fit for the systems of secure communications. When a user requests a file, the system guards the integrity of the file by initiating an integrity check for both the user requested file and the system file to compare for the researcher system. Empirical analysis shows for its use, improved data availability, increased security, improved resistance to attacks significantly, and decreased storage costs that were the result of the use of a pure blockchain-based system. Ultimately, the study shows there are positive advancements in the use of IPFS and blockchain systems to incorporate greater modifications to the systems for the distribution and sharing content securely.

Keywords— Blockchain, IPFS, Secure File Sharing, Smart Contracts, Data Integrity

I. Introduction

In the domain of digital data exchange, the need for safe file exchange systems rapidly grows with the growing reliance on them for digital systems. Education, healthcare, and financial sectors are some of the most impacted. Many industries are facing increasing demand for secure and reliable pathways for digital file transfer and storage due to the growing dependence of their activities on digital systems. Traditional systems of digital storage are inherently vulnerable to cyberattacks because they are organized around a central server [3], [7]. The so-called central point of failure leaves the entire system liable to unauthorized access, manipulation, or even complete loss of the stored data, thus eroding customer confidence. In response to these challenges, systems are needed that are neither effective, nor dependent on a central authority.

One such system rests on the combination of blockchain and IPFS, the InterPlanetary File System. Blockchain is a highly secure and transparent, immutable log of transactions that eliminates the need for external authority, while IPFS is a system of distributed files organized through a peer-to-peer network [1], [3], [5]. Regardless of that, the act of storing files directly on the blockchain is neither economical nor efficient.

In light of the aforementioned challenges, this study presents a system utilizing the IPFS for the distributed storage of files in such a way that a file-oriented hash is created for the purpose of content addressing [4]. In support thereof, the aforementioned hashes are recorded on the blockchain to uphold the integrity and authenticity of the files [1], [3]. This system eliminates the external storage and the verification interdependence for large scale blockchain systems preserving immutability.

The combination of blockchain technology and IPFS has solved problems surrounding security, data safety, and user confidence. It has also made data tracking and file control more manageable [1], [3]. When using a blockchain, users do not depend on a central authority, to verify everything. In addition, the proposed system provides users with increased control of their digital assets, by considerably minimizing storage duplication. Thus, the system is secure, cost-effective and storage-efficient. It significantly reduces bandwidth use by eliminating

redundant file storage allowing for large scale deployments.

II. Methodology

A. System Architecuture

The proposed system architecture attempts to modify components of integrity and central data storage by decentralizing data integrity and storage. This hybrid model of decentralized architecture, provides large scale data access and integrity proofing, and eliminates the storage of raw data on the blockchain [5].

The system comprises of four primary integrated layers such as:

1. User Interface for file interaction: IPS (via Pinata) for decentralized storage. The system comprises of four primary integrated layers such as: User Interface for file interaction. The system uses this layer to provide users with access to the decentralized network. It is built using react.js and aims to provide a hassle free experience. It regulates the boundaries of the users' local environment, defines the logic for the file selection process, and activates the hash command.

2. Decentralized Storage Layer using IPFS: IPFS, the InterPlanetary File System, is used to fragment the files and distribute the layers across the peer-to-peer network [5]. Pinata is used here as a pinning service for IPFS, ensuring that files remain in the IPFS system and are not subject to potential deletions originated within the IPFS environment.

3. Blockchain Integrity Layer: This Ethereum network acts as a "unbreakable anchor" for the system. The Ethereum blockchain, used here, does not need to retain the actual data. It retains only the unique Content Identifier (CID) generated by IPFS [1]. The logic for preserving these hashed as well as the access criteria, are described by smart contracts that the user created on the state of the blockchain, using the Remix IDE. Therefore, the data is permanent.

4. Security & Gateway Layer: MetaMask offers users a secure layer of identity to support transactions and payment for the network [1]. A browser extension for Ethereum blockchain operates as a wallet and secure authentication gateway. Therefore, users are supported as they safely upload documents and cover network costs to register file hashes.

B. Workflow Implementation

The practical workflow starting point is as follows: [5]

Start

↓

Select File

↓

Upload to IPFS (Pinata)

↓

Generate Hash (CID)

↓

Store Hash on Blockchain (Smart Contract by using Remix)

↓

Retrieve File using Hash

↓

Verify Integrity

↓

End

Once a user selects a file through the interface, it is uploaded to the IPFS network via Pinata. The file is then split into multiple pieces, cryptographically secured [4], and distributed across a self-healing network. The IPFS network also creates and assigns a unique content identifier (CID) to the file [5]. The CID is then uploaded to a blockchain as a permanent record via a smart contract created in Remix. Users retrieve files from the IPFS by using the CID, and recompute the file's cryptographic hash. The computed hash is then validated by a comparison to the CID that is stored on the blockchain to confirm the data integrity [4].

C. Working Mechanism

1. *File Upload:* Users select a file that is to be uploaded to the IPFS network via Pinata.
2. *Hash Generation:* The IPFS network creates a unique content Identifier CID that is derived from the file content [5].
3. *Blockchain Storage:* The CID is stored in a smart contract on the blockchain via Remix [1].
4. *File Retrieval:* Users retrieve files using the CID or hash.
5. *Integrity Verification:* The user validates the file to determined if it matches the stored file as it was originally uploaded [4].

D. Equations

The system relies on cryptographic hash functions, that can be expressed as follows:

Hash Function Representation

$$CID = H(file) \text{ [4]}$$

Integrity Verification

$$H(file_{retrieved}) = H(file_{original}) \text{ [4]}$$

Tampering Detection

$$\text{If } H(file_{new}) \neq H(file_{stored}), \text{ then the file is modified}$$

Block Storage Representation

$$Block = \{CID, Timestamp, Previous_Hash\} \text{ [1], [4]}$$

Encryption (optional Enhancement)

$$E(file) = Enc(file, key) \text{ [4]}$$

$$D(file) = Dec(E(file, key)$$

Where H represents the hash function, and the output is fixed-length unique identifier for the file [4]. Formal representations like these enhance the technical rigor of the system and provide a foundation to its substantiation.

E. Tools and Technologies used

This system is integrated with technologies needed to enhance performance and efficiency. For decentralized file storage, the system uses IPFS [1], [5].

File availability on IPFS is maintained through Pinata. For blockchain operations, Ethereum is used, and smart contracts are created via the Remix IDE. These smart contracts store safety file IDs [1]. Hence, the data remains safe against alterations.

The user interface was developed using React Js, thus simplifying usage. MetaMask is used as a wallet to securely connect the user interface with the blockchain network.

III. Results

A. *The implemented system successfully demonstrates:*

1. Compared with traditional systems, this system provides better security, reliability, and transparency [1], [3].

2. Files are stored in IPFS in a decentralized manner; therefore, there is no single point of failure [5]. Each file receives a unique CID, which helps identify it and avoids storing duplicate files.

3. CIDs cannot be changed once they are stored on the blockchain. This prevents unauthorized changes and makes it easier to track the data [4].

B. *Performance Observations*

1. *Integrity Validation:* The system removes the risk of a single point of failure and is robust against cyber-attacks, such as hacking or data tampering. It can easily detect whether any file has been changed [4].

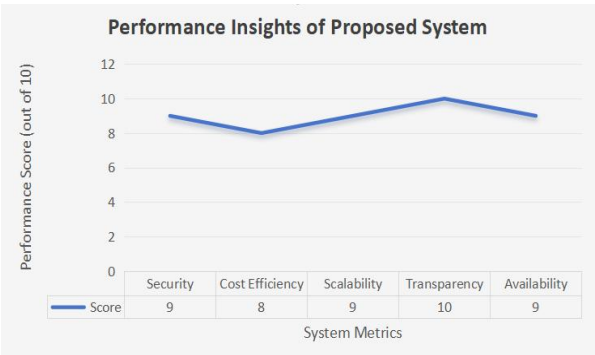

Fig. 1. *Performance Insights of the Proposed System*.

2. Efficiency in Storage: There can be a small delay because blockchain transactions take time to confirm, but overall, the system performs well. By separating the storage layer (IPFS) for the verification later (Blockchain), it handles large files efficiently also benefits such as security and transparency are more important than this minor delay.

3. Cyber-Attack Resistance: Based on empirical assessment, the decentralized design offers protection against typical risks such as hacking, server compromises, and data breaches.

4. Cost Management: This hybrid solution offers greater security than conventional centralized systems while drastically lowering storage costs when compared to entirely blockchain-based storage.

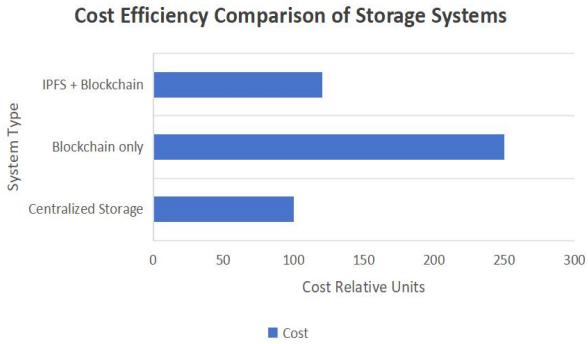

Fig. 2. *Cost Efficiency Comparison of Storage Systems*.

IV. Discussion

A. *Overview of the Proposed System*

The innovative design of the secure file-sharing system proves the power of technologies, as blockchain and IPFS, to solve most of the important limitations regarding the traditional centralized structures [1], [3]. Instead of directly storing files on the blockchain, this system separates the storage and verification layers such that large files are efficiently handled by IPFS and lightweight [5], immutable records of file hashes are maintained by the blockchain. This hybrid approach significantly improves performance while enhancing data integrity and security [3].

B. *Advantages of the System*

One of the key benefits of this system is its security feature. Each file uploaded to the IPFS is assigned a distinct content identifier; thus any alteration to the file results in a completely new hash. By storing this hash on the blockchain, immutability is guaranteed, making it easy to detect any changes or unauthorized modifications [4].

Another significant advantage is decentralization, as the inherent nature of both blockchain and IPFS removes single-point failure, thereby enhancing resilience against cyberattacks, such as server compromises and data breaches, which in turn makes the system more dependable [3], [6], [7].

The system is transparent because anyone can verify the data stored on the blockchain. In addition, IPFS improves availability by storing files across multiple nodes instead of just one location.

C. *Figures and Tables*

TABLE I. Comparison with traditional file sharing systems [3]

Feature	Traditional System	Proposed System-
Storage Type	Centralized	Decentralized(IPFS)
Security	Vulnerable	Highly Secure
Data Integrity	Not Guaranteed	Guaranteed via Hash
Single Point of Failure	Yes	No
Transparency	Limited	High

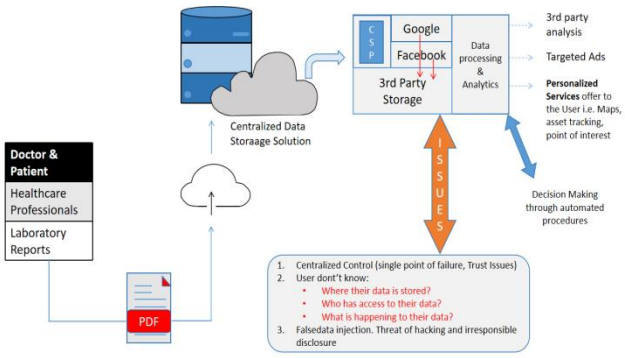

Fig. 1. *Challenges in storage using a centralized approach.*

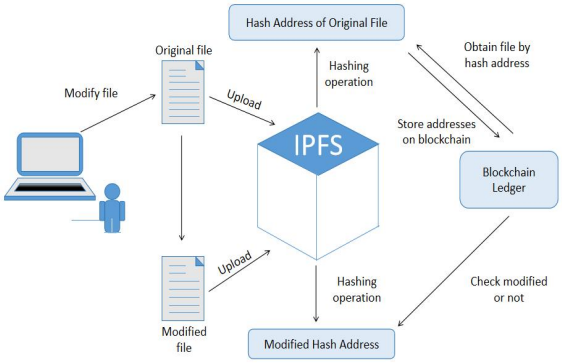

Fig. 2. *System architecture showing interactions between users, IPFS (Pinata), and blockchain networks.* [1], [5]

D. Limitations and Challenges

While the proposed decentralized framework offers significant improvements over traditional centralized architectures, several technical and economic challenges must be addressed for large-scale adoption.

One major problem is the cost of blockchain transactions, especially on public networks, where fees can be high [3] such as a transaction fee known as gas, recording a new file hash. This makes it difficult to use on a large scale as these fees can fluctuate wildly based on network congestion, which sometimes make the systems expensive for frequent small-scale file sharing. These cumulate costs can hinder the financial feasibility for enterprise level deployments.

The system introduces a trade-off between absolute security and operational speed however, systems are bound by latency which is another limitation as blockchain operations may take time to confirm transactions, which slows down the process compared with conventional systems. A user must wait for the blockchain network to mine and validate the block containing their file's CID before the upload is considered "officially" recorded.

Moreover, even though IPFS offers decentralized storage, files might become inaccessible over time if they are not properly pinned using services such as Pinata which has 24/7 data availability [5].

E. Applications of the proposed system

This proposed system can be applied in multiple domains such as:

1. Healthcare, which allows safe storage and sharing of patients' data with privacy and integrity [2], [6], [7].

2. Finance,which makes for safe sharing of documents and the creation of systems to prevent fraud [3].

3. Education, which allows safe sharing of students' records, including academic transcripts and certificates [3].

This system can be applied in many disciplines, including supply chain management, digital identity, and other disciplines, where data integrity and transparency are of utmost importance.

V. Conclusion

A. Work Summary

This research shows a method to offer safety in decentralized file sharing systems by designing a system that offers secure file distribution while preserving file integrity through decentralized systems, IPFS, and blockchain hash storage [1], [3]. This modular system was implemented by using Pinata and Remix.

B. Key Contributions

The result of this research is the construction of a secure and stable modular system that is resistant to breaches and making a system using blockchain and IPFS, which reduces the manipulation and unauthorized access of data, and also provides user trust during a data breach.

C. Practical Importance

The secure system proposed in this study can also be utilized in a variety of other fields, such as finance for the secure exchange of sensitive information; healthcare for the protection of sensitive information; and education for the protection of sensitive information. This offers security and integrity to a wide range of data in a variety of fields.

D. Future prospects

In the future this system is anticipated to provide enhancements in the forms of large scale and cross platform decentralized system, and faster transactions that provide a greater degree of security.

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